

Remote Energy Monitoring

BEng(Hons) MSc PhD(Cantab) GMCOP CEng MCIBSE

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Many organisations intend to upgrade to a remote energy monitoring system, to understand energy use and improve service performance across their globally distributed estates. Having a better understanding of energy use should allow strategies and programmes to be developed holistically at a local level. The following requirements provide guidance on how to inter-network with such an application.

1 Utility Metering

The tables below compare the various types of utility meters available.

Table 1: Internet-of-Things (IoT) meter

Advantages	Disadvantages
No requirement for inter-networking gateway	Expensive
Meter counts energy consumption	Device testing requirement to validate performance
Potentially more descriptive information payload.	

Table 2: Open building systems communications protocol meter

Advantages	Disadvantages
Meter counts energy consumption	Insufficient security features for exposure to the Wide Area Network (WAN).
Potentially more descriptive information payload.	May utilise broadcast protocols.

Table 3: Pulse meter

Advantages	Disadvantages
Relatively cheap	Requires a pulse counting device
Simple	Minimal information payload

2 Inter-networking

It is important that any inter-networking gateways be hosted on a dedicated Virtual Private Network, with layer 3 separation from other building management system networks. Otherwise, exposure of such inter-networking gateways to the Wide Area Network would pose an unacceptable cybersecurity risk.

Table 4: Inter-Networking Gateway

Inter-Networking Standard	Justification
802.1x	Certificate based network access managed by a Remote Authentication Dial-In User Service (RADIUS)
IP version 6	Unique, lower cost and permanent public Internet Protocol (IP) addressing, with native support for IPSec encryption of entire data packets in transit.
TLS version 1.3	Most recent release of Transport Layer Security (TLS) protocol, for encryption of data packets transport layer in transit.
LDAP	Lightweight Directory Access Protocol (LDAP) is used for accessing and maintaining distributed directory information services over an Internet Protocol (IP) network.

hoshi.property prefers telemetry over Secure Message Queue Telemetry Transport (MQTTS) for the following reasons:

- All inter-networking communications managed from a single endpoint;
- Lightweight telemetry;
- No packet loss, even through interruptions to network connectivity.

A MQTTS message broker is available at the URL: `mqttp.hoshi.property` and may subscribe to secure telemetry, given recognised 802.1X and TLS certificates. Each inter-networking gateway shall be allocated a Globally Unique Identifier (GUID), serving as the message brokers topic. Customers may link available Guid's to service addresses through the `hoshi.property` user interface.

`mqttp.hoshi.property` only accepts telemetry that strictly follows the schema:

`../schemas/utilisation.json`